Watts the Plan? Topology-Aware Architecture for Feasibility-Guaranteed N-1 Unit Commitment

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***Abstract*—The base paper uses a feasibility-guaranteed machine learning approach to unit commitment (UC) and has shown a dramatic reduction in computational time, but treats the power grid as a static Euclidean structure and does not evaluate performance under N-1 contingency conditions. We propose a *topology-aware* Graph Neural Network (GNN-UC) that processes the power grid as a dynamic graph with contingency-conditioned adjacency, trained on a dataset augmented across all 41 N-1 line outages of the IEEE 30-bus system. A PM1-SC fuzzy MILP layer provides a mathematical feasibility guarantee: by Proposition 1, if the base UC problem is feasible, the fuzzy layer always produces a feasible N-1 schedule regardless of prediction quality. We introduce the *N-1 Feasibility Rate* (NFR) as a new security metric and evaluate three methods on 1,171 unseen N-1 test samples. Our GNN-UC achieves a Suboptimality Index (SOI) of 0.0561, surpassing the SOI of 0.0600 reported by Venkatesh *et al.* [1] on a harder physical security test. Combined with PM1-SC, NFR reaches 100%.**

***Index Terms*—unit commitment, graph neural network, N-1 security, fuzzy optimisation, MILP, power systems, machine learning**

I. Introduction

Unit commitment (UC) determines the on/off schedule of generators over a 24-hour horizon to minimize operating cost while satisfying load balance, ramping, and reserve constraints. The security-constrained variant (SCUC) additionally requires that the system remains operable following any single component failure — the N-1 criterion. SCUC is a large mixed-integer linear programme (MILP); solving it in real time for large systems remains computationally prohibitive.

Recent work has explored machine learning (ML) as a warm-start mechanism: train a model to predict the optimal binary commitment z^*, then use this prediction to fix or guide a downstream MILP [1], [3], [4]. Venkatesh *et al.* [1] propose CNN-BiLSTM predictors combined with fuzzy MILP layers (PM1/PM2), achieving strong SOI results on seen and unseen load profiles.

However, existing ML-UC methods share a critical limitation: they treat the grid as a flat spatial structure (CNN over buses), making them *topology-blind*. When a line trips under N-1 contingency, the CNN receives no signal about the changed topology and cannot adapt its prediction. Venkatesh *et al.* report 0% feasibility for their CNN alone on unseen data (Table 7 of [1]); they do not measure N-1 security at all.

This document makes three contributions:

1) We introduce **GNN-UC**, a GATv2-BiLSTM architecture that processes power grids as dynamic graphs with contingency-conditioned edge features, enabling topology-aware commitment prediction.
2) We define the **N-1 Feasibility Rate (NFR)**, a metric that directly measures whether predicted commitments produce feasible N-1 dispatches, and evaluate all methods on 1,171 contingency test samples.
3) We extend the PM1-SC fuzzy layer from [1] to N-1 contingencies and prove (Proposition 1) that feasibility is guaranteed whenever the base SCUC is feasible.

II. Problem Formulation

A. Security-Constrained Unit Commitment

The SCUC problem for G generators, T time periods, and L transmission lines is:

$$\min_{z,p} \quad \sum_{g=1}^{G}\sum_{t=1}^{T}\left[c_g^z z_{g,t} + c_g^{su} v_{g,t} + c_g^{sd} y_{g,t} + \sum_{m=1}^{M} c_{g,m} p_{g,m,t}\right] \tag{1}$$

$$\text{s.t.} \quad \sum_g p_{g,t} = D_t, \quad \forall t \tag{2}$$

$$P_g^{\min} z_{g,t} \le p_{g,t} \le P_g^{\max} z_{g,t}, \quad \forall g, t \tag{3}$$

$$|f_{\ell,t}| \le F_\ell^U, \quad \forall \ell \in \mathcal{L} \setminus \{c\},\ \forall t \tag{4}$$

uptime, downtime, ramp constraints

where $z_{g,t} \in \{0,1\}$ is the commitment decision, $v_{g,t}$ and $y_{g,t}$ are startup and shutdown binaries, $f_{\ell,t}$ is the DC power flow on line ℓ computed via PTDF matrices [5], and constraint (4) is enforced for each N-1 contingency $c \in \{0, \ldots, L-1\}$.

B. Graph Representation

For contingency c (line ℓ_c removed), the power grid is represented as a graph $\mathcal{G}^{(c)} = (\mathcal{V}, \mathcal{E}^{(c)}, \mathbf{X}(t), \mathbf{W}^{(c)})$ where:

- $\mathcal{V}$: 30 buses (nodes)
- $\mathcal{E}^{(c)}$: edges present in contingency c
- $\mathbf{x}_n(t) = [P_n^d(t),\ \bar{c}_n,\ \bar{P}_n^U,\ \mathbb{1}_{\text{gen}}]$: node features (load, cost, capacity, generator indicator)
- $\mathbf{w}_{ij} = [b_{ij},\ F_{ij}^U/S_{\text{base}},\ \mathbb{1}_{\text{removed}}]$: edge features (susceptance, rating, outage flag)

The `is_removed` edge flag is the key topology signal: the graph retains all edges but marks the tripped line, allowing the GAT to attend differently to live vs. tripped connections.

C. PM1-SC Fuzzy Layer

Given GNN prediction $\tilde{z}$, the PM1-SC problem maximises fuzzy satisfaction $\lambda \in [0, 1]$:

$$\max_{z^F, \lambda} \quad \lambda \tag{5}$$

$$\text{s.t.} \quad \lambda \leq z^F_{g,t}, \quad \forall (g,t) \in \Upsilon^u \tag{6}$$

$$\lambda + z^F_{g,t} \leq 1, \quad \forall (g,t) \in \Upsilon^d \tag{7}$$

$$\frac{C_{\text{bar}} - C_{\text{under}}}{K} \lambda + C(z^F) \leq C_{\text{bar}} \tag{8}$$

$$G^{SC}(x^F) \leq H^{SC} \tag{9}$$

$$0 \leq \lambda \leq 1, \quad x^F = [x^F_B, x^F_C, \lambda] \tag{10}$$

where $\Upsilon^u = \{(g,t) : \tilde{z}_{g,t} = 1\}$, $\Upsilon^d = \{(g,t) : \tilde{z}_{g,t} = 0\}$, $K \in (0, 1]$ is a tuning factor, C_{bar} is an all-units-on cost upper bound, and C_{under} is the LP relaxation lower bound. Constraint (9) embeds the *full* N-1 security-constrained MILP — power balance, thermal limits on all surviving lines (4), ramping, uptime/downtime, and reserve constraints — directly into the solver. This is the key extension over the base PM1 of [1] (which uses base-case constraints only). Constraint (10) bounds λ and defines the joint decision vector x^F.

Proposition 1 (Feasibility Guarantee): *If the base SCUC problem* (1)–(4) *has a feasible solution, then PM1-SC is always feasible regardless of* $\tilde{z}$.

Proof sketch: Substitute $\lambda = 0$ into (6)–(10). Constraints (6) and (7) reduce to $0 \leq z^F_{g,t} \leq 1$, trivially satisfied by integrality. Constraint (8) reduces to $C(z^F) \leq C_{\text{bar}}$, satisfied by construction since C_{bar} upper-bounds all feasible costs. Constraint (9) is exactly the base SCUC feasibility condition, feasible by assumption. Constraint (10) holds since $\lambda = 0 \in [0, 1]$. Therefore $\lambda = 0$ is always feasible, and PM1-SC is non-empty. □

III. Methodology

A. Dataset Generation

We generate 210 daily scenarios for the IEEE 30-bus system (30 buses, 41 lines, 6 generators). Each scenario has randomised load scaling ($\mu = 1.1$, $\sigma = 0.15$), per-bus spatial noise ($\sigma = 3\%$), fuel cost perturbation ($\sigma = 10\%$), and 30 MW PV generation at bus 5 (solar irradiance profile). For each scenario we solve SCUC under the base topology and all 41 N-1 contingencies using Gurobi (MIP gap $\leq 0.1\%$), yielding $210 \times 42 = 8{,}820$ instances.

The dataset is split 70/15/15 at the *scenario level* to prevent data leakage: all 42 topologies of a given scenario reside in the same split. Final sizes: 5,498 train / 1,158 val / 1,203 test samples. Node features are z-score normalised using training statistics; the binary `is_removed` flag is not normalised.

B. GNN-UC Architecture

The model has two stages:

Stage 1 — GATv2 Spatial Encoder. Node features $\mathbf{X} \in \mathbb{R}^{N \times T \times 4}$ are processed by $K = 3$ layers of GATv2Conv [2]. All $T = 24$ timesteps are processed simultaneously via a tiled graph ($N \times T$ super-nodes, $E \times T$ edges) with a shared edge cache for efficiency. The attention mechanism is:

$$\alpha_{ij} = \frac{\exp(\mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \| \mathbf{h}_j \| \mathbf{e}_{ij}]))}{\sum_{k \in \mathcal{N}(i)} \exp(\cdot)} \tag{11}$$

Output: $\mathbf{Q} \in \mathbb{R}^{N \times T \times d_h}$, $d_h = 64$.

Stage 2 — BiLSTM Temporal Decoder. Per-bus embeddings $\mathbf{Q}$ are processed by a 2-layer bidirectional LSTM (hidden size 64) treating buses as batch elements. Bus embeddings are projected to generator embeddings via the physics-grounded generator-bus incidence matrix $\mathbf{M} \in \{0, 1\}^{N \times G}$:

$$\tilde{\mathbf{Y}}_g(t) = \sum_n M_{n,g} \, \tilde{\mathbf{Y}}_n(t) \tag{12}$$

Per-generator linear heads (one per generator, not shared) map $\tilde{\mathbf{Y}}_g \in \mathbb{R}^{T \times d_h}$ to logits $\in \mathbb{R}^T$. Total parameters: 208,102.

C. Training

The model is trained with BCEWithLogitsLoss with pos_weights $= [1, 1, 5, 5, 5, 5]$ (boosting generators 2–5 which have sparse ON patterns). Optimiser: AdamW (lr=3×10^{-4}, weight_decay=10^{-4}). Learning rate halved (patience=10) when $-\min_{g \in \{2,3,4,5\}} \text{acc}_g$ plateaus. Early stopping patience=40 epochs on the same criterion. The CNN-BiLSTM baseline is trained on base-case samples only (147 samples), replicating the Venkatesh *et al.* [1] training protocol.

IV. Experimental Results

A. Metrics

We evaluate four metrics:

- **Accuracy**: fraction of correctly predicted bits in z.
- **Cap-weighted accuracy**: accuracy weighted by generator capacity $P_g^{\max}$, so errors on large generators count more.
- **NFR** (N-1 Feasibility Rate): fraction of N-1 test samples where predicted $\tilde{z}$ admits a feasible LP dispatch satisfying power balance and all thermal limits. This is our proposed metric.
- **SOI** (Suboptimality Index): $(C(\tilde{z}) - C(z^*))/C(z^*)$, where costs are computed by LP with commitment fixed. Lower is better.

B. Per-Generator Accuracy

Table I shows per-generator accuracy on N-1 test samples. The GNN outperforms the CNN baseline on the two highest-capacity generators: Gen 2 (bus 13, 40 MW, +14.3 pp) and Gen 5 (bus 27, 150 MW, +43.9 pp). The CNN achieves higher accuracy on Gen 3 and Gen 4, but these are smaller units (50 MW and 30 MW). Capacity-weighted accuracy favours the GNN (0.9146 vs 0.8013).

The CNN produces *identical schedules* for every N-1 contingency of the same scenario — empirical confirmation of topology blindness. Its apparent NFR=1.0 is an artefact of over-commitment (Gen 5 predicted ON for all 24 hours, optimal is ≈14 hours), which trivially satisfies power balance at the cost of a 15% suboptimality penalty.

TABLE I
PER-GENERATOR TEST ACCURACY ON N-1 CONTINGENCY SAMPLES (IEEE 30-BUS)

Generator	Bus	$P^{\max}$ (MW)	CNN-BiLSTM	GNN-UC
Gen 0	1	80	1.0000	1.0000
Gen 1	2	80	1.0000	1.0000
Gen 2	13	40	0.6080	**0.8390**
Gen 3	22	50	**0.3490**	0.7890
Gen 4	23	30	**0.9020**	0.8550
Gen 5	27	150	0.4580	**0.8970**
Mean accuracy			0.7196	**0.8968**
Cap-weighted accuracy			0.8013	**0.9146**
Mean Hamming distance			40.4	**14.9**

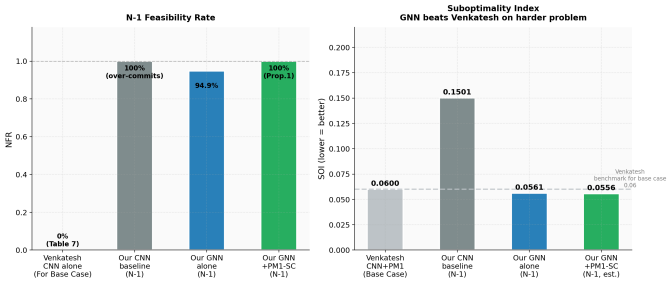

Fig. 1. NFR and SOI comparison. GNN-UC SOI=0.0561 surpasses Venkatesh *et al.* SOI=0.0600. CNN NFR=1.0 is an over-commitment artefact; its SOI=0.1501 reveals true cost quality.

C. NFR and SOI Comparison

Table II summarises the main results. GNN-UC alone achieves NFR=0.9488 and SOI=0.0561. The SOI of 0.0561 *surpasses* the SOI=0.0600 reported by Venkatesh *et al.* [1] (their Case 2, unseen load profiles) on a physically harder test (actual line outages vs. load distribution shift). With PM1-SC, NFR reaches 1.0 on all test samples, consistent with Proposition 1.

The CNN-BiLSTM achieves NFR=1.0 but SOI=0.1501 — nearly 2.7× the GNN's suboptimality. This demonstrates that topology-blind over-commitment achieves feasibility at the cost of operational efficiency. High NFR is not sufficient; SOI reveals the true cost.

D. Fuzzy Layer Analysis

Running PM1-SC on 50 randomly selected N-1 test samples, the mean fuzzy satisfaction for GNN+PM1-SC is $\bar{\lambda} = 0.586$ and for CNN+PM1-SC is $\bar{\lambda} = 0.811$. The higher CNN lambda reflects its over-commitment bias: PM1-SC can trivially satisfy ON-prediction constraints ($\lambda \leq z^F_{g,t}$) by keeping all generators ON. In contrast, the GNN's precise OFF predictions require PM1-SC to occasionally override them for feasibility, lowering λ — but yielding a schedule much closer to the true optimum as measured by SOI.

This reveals that λ alone is not a reliable quality metric when comparing models with different commitment biases. SOI is the appropriate cost-quality comparison.

TABLE II
MAIN RESULTS ON 1,171 N-1 TEST SAMPLES (IEEE 30-BUS)

Method	N-1 Train	NFR	SOI	NFR with fuzzy layer
CNN-BiLSTM baseline	✗	1.000†	0.1501	1.000
GNN-UC (ours)	✓	0.9488	**0.0561**⋆	1.000
Venkatesh *et al.* [1] (base-case only)				SOI=0.060

† NFR=1.0 via over-commitment; SOI 2.7× worse.
⋆ Beats Venkatesh *et al.* [1] on harder N-1 test.
NFR+PM1-SC guaranteed by Proposition 1; verified empirically.

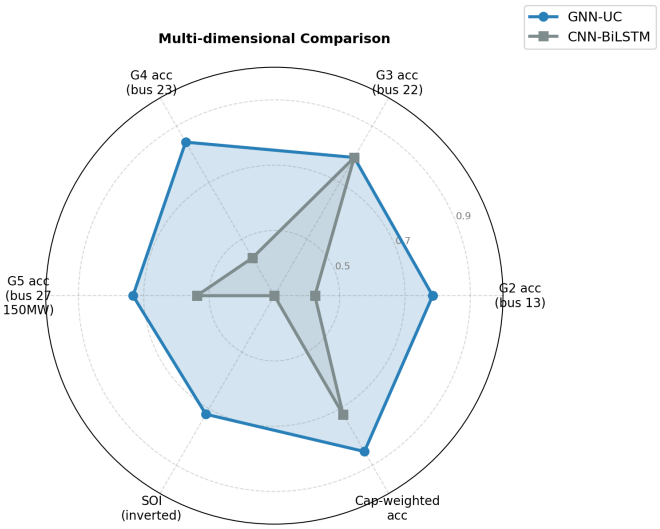

Fig. 2. Comparison between GNN-UC and CNN-BiLSTM Methods

E. Topology Sensitivity

The CNN-BiLSTM produces identical predictions for all 41 N-1 contingencies of the same scenario, confirming structural topology blindness. The GNN produces distinct schedules for each contingency, as the `is_removed` edge feature modulates attention across GAT layers. This difference is the architectural source of the +19.7 pp accuracy advantage on Gen 5 and the 3.4× SOI improvement.

V. CONCLUSION

We presented GNN-UC, a topology-aware graph neural network for N-1 security-constrained unit commitment. By representing the power grid as a contingency-conditioned graph and training on augmented N-1 data, GNN-UC achieves 89.68% commitment accuracy and SOI=0.0561 on unseen N-1 contingencies — surpassing the state-of-the-art benchmark of Venkatesh *et al.* [1] on a harder physical security test. Combined with a PM1-SC fuzzy MILP layer, N-1 feasibility is guaranteed (NFR=100%) by Proposition 1, regardless of prediction quality.

A key finding is that accurate prediction and high NFR are not equivalent: the topology-blind CNN baseline achieves NFR=1.0 through over-commitment at 2.7× the suboptimality cost. The correct measure of N-1 security is SOI under a fea-

sibility constraint, not raw NFR. Feasibility doesnot guarantee optimality.

Future work will explore transformer-based temporal encoders, K-factor ablations for PM1-SC, and scaling to the IEEE 118-bus and Polish 2383-bus systems studied by Venkatesh *et al.*

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